



Optimization Homework 1

SF1811

January 9, 2026

Lucas Frykman, Jonathan Tadesse

Student personal numbers: 021012-7650, 970729-4311

Student Emails: lfrykman@kth.se, jtadesse@kth.se

AI level 1: Used chatgpt to parse numerical results in to latex.

The intent of this homework assignment is to use the simplex method for solving a linear program in standard form,

$$\begin{aligned} \text{(LP)} \quad & \min_{x \in \mathbb{R}^n} c^T x \\ & \text{subject to } Ax = b, \\ & x \geq 0. \end{aligned}$$

There are Matlab functions `simplex.m` and `polyplot.m` available in Canvas to assist you.

Exercise 1

The simplex method is an iterative method. Its aim is to find the minimum of functions of the form $z(x) = C^T x$ with a constraint $Ax = b$. Here's a brief description of each step of the method.

Select the initial basic index tuple

We start by selecting linearly independent columns of A to form a new matrix A_β which we call the basic matrix. The basic index tuple $\beta = (\beta_1, \beta_2, \dots, \beta_m)$ are the indices of the basic variables we have selected from x in to x_β corresponding to the column indices we have selected for the basic matrix. After this we define our non basic matrix and non basic variable A_ν and x_ν where the non basic index tuple $\nu = (\nu_1, \nu_2, \dots, \nu_{n-m})$ are all the indices not included in β . Where we set $x_\nu = 0$.

Step 1:

Solve $A_\beta \bar{b} = b$ for $\bar{b}$. This is solvable with multiplying with A_β^{-1} . If $\bar{b} \geq 0$ this gives us the initial basic feasible solution for x_β . Our goal is to determine whether this solution is optimal or if we can decide on a better choice of basic variables.

Step 2:

Solve $A_\beta^T y = c_\beta$ for simplex multipliers y . (Similarly as before c_β are the corresponding coefficients to x_β)

Step 3:

Calculate $r_\nu^T = c_\nu^T - y^T A_\nu$ to find the reduced cost with respect to the non basic variables. Observe that this value is obtained from the objective value function $z = y^T b + (c_\nu^T - y^T A_\nu)x_\nu$

This measures the rate the objective value function improves if our choice of non-basic variables have been increased from zero.

Step 4: stopping criterion

if $r_\nu \geq 0$ **then**

then there's no non basic variables that can perform better than the basic variables we have chosen. i.e the basic feasible solution we found in this iteration is optimal and we can stop (any further increase to will increase the objective function).

return $x_\beta = \bar{b}$, $x_\nu = 0$

else

Step 5: Select entering variable

Choose index q such that $r_{\nu_q} < 0$. By dantzig's rule we choose q from the most negative component of r_ν .

Set $k = \nu_q$. We want to try to find a better BFS so we'll let $x_k = x_{\nu_q}$ be a new basic variable.

Step 6:

Let $x_{\nu_q} = t$ while we keep other non basic variables at 0. The cost function will become $z = \bar{z} + r_\nu^T x_\nu = \bar{z} + tr_{\nu_q}^T$ along with the constraint $A_\beta x_\beta + ta_k - b = 0$. We can factor out A_β from this equation since the basic matrix is invertible so we obtain that $A_\beta(x_\beta + t\bar{a}_k - \bar{b}) = x_\beta + t\bar{a}_k - \bar{b} = 0$.

so in this step we need to solve

$$A_\beta \bar{a}_k = a_k$$

From this we get the following result

$$x_\beta = \bar{b} - t\bar{a}_k$$

If $\bar{a}_k \leq 0$ then t can increase to infinity without violating feasibility and the cost z will go to $-\infty$ and no optimal solution exists. But if at least one positive element exists in $\bar{a}_k$ then we want to calculate $t_{\max}$ to see what's the farthest we can go to decrease our cost function $z(t) = \bar{z} + tr_{\nu_q}^T$ without violating feasibility. This means we can obtain a new BFS that's better than what we currently have.

Step 7: Ratio test (select leaving variable)

As stated in step 6, to ensure the feasibility condition $x_\beta \geq 0$ we want to know the maximum step size $t_{\max}$. This tells us how much t can increase before a variable in x_β goes to zero. We want to remove the first basic variable that does this. Lets call the index of the leaving variable p . When calculating $t_{\max}$ we ensure $\bar{a}_{ik} > 0$ and identify index p :

$$t_{\max} = \min_{i: \bar{a}_{ik} > 0} \left(\frac{\bar{b}_i}{\bar{a}_{ik}} \right) = \frac{\bar{b}_p}{\bar{a}_{pk}}$$

Step 8: ν_q and β_p change place when we update our index tuples β, ν then repeat from step

1

end if

Exercise 2

Lucas's personal number was used to get the data from `hw1data`. Our matrices are given by

$$A = \begin{pmatrix} -0.9771 & 0.9951 & 1.0000 & 0 & 0 & 0 \\ -0.0085 & 0.9947 & 0 & 1.0000 & 0 & 0 \\ 1.9312 & 2.9730 & 0 & 0 & 1.0000 & 0 \\ 1.9425 & -1.0206 & 0 & 0 & 0 & 1.0000 \end{pmatrix}, b = \begin{pmatrix} 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix}, c = \begin{pmatrix} -1.0324 \\ -2.9633 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Where we have to solve the LP:

$$\begin{aligned}
 \min \quad & z = -1.0324 x_1 - 2.9633 x_2 \\
 \text{subject to} \quad & -0.9771 x_1 + 0.9951 x_2 + x_3 = 1.9526, \\
 & -0.0085 x_1 + 0.9947 x_2 + x_4 = 2.9920, \\
 & 1.9312 x_1 + 2.9730 x_2 + x_5 = 14.0105, \\
 & 1.9425 x_1 - 1.0206 x_2 + x_6 = 6.0423, \\
 & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.
 \end{aligned}$$

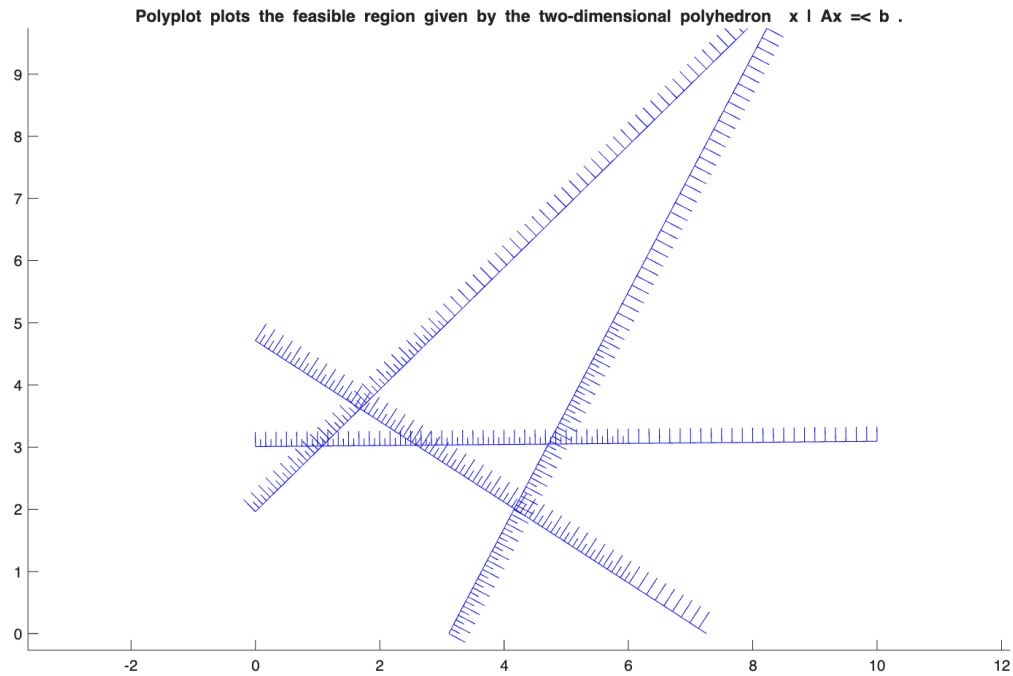


Figure 1: feasible region

Exercise 3

Iteration 1

Step 1: Basic Solution

We have that $A_\beta = I$ with linearly independent columns where the basic index tuple is $\beta = (3 \ 4 \ 5 \ 6)$

$$\text{We first solve } A_\beta \bar{b} = b \Leftrightarrow \bar{b} = \begin{pmatrix} 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix} \Rightarrow x^{(1)} = \begin{pmatrix} 0 \\ 0 \\ 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix}$$

Step 2: Simplex multipliers

$$\text{Solve } A_\beta^T y = c_\beta \text{ for } y \Leftrightarrow I^T y = y = c_\beta = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Step 3: Reduced costs

$$\text{Calculate } r_\nu^T = c_\nu^T - y^T A_\nu = c_\nu^T - 0^T A_\nu = c_\nu = \begin{pmatrix} -1.0324 \\ -2.9633 \end{pmatrix}$$

Step 4: stopping criterion

We see that $r_\nu < 0$ so the BFS isn't optimal

Step 5: Select entering variable

Choose index $q = 2$ since $r_{\nu_2} < 0$ and $r_{\nu_2} = -2.96 = \min_{\nu_i \in \nu} r_{\nu_i}$ is the most negative value of r_ν .

So set $k = \nu_2$

Step 6:

$$\text{Solve } A_\beta \bar{a}_k = a_k \text{ for } \bar{a}_k \Leftrightarrow I \bar{a}_k = \begin{pmatrix} 0.9951 \\ 0.9947 \\ 2.9730 \\ -1.0206 \end{pmatrix}$$

$\bar{a}_k$ holds positive elements so we're able to get a new BFS and continue to step 7

Step 7: Ratio test (select leaving variable)

Calculate t_{max} and identify index p :

$$t_{max} = \min_{i: \bar{a}_{ik} > 0} \left(\frac{\bar{b}_i}{\bar{a}_{ik}} \right) = \frac{\bar{b}_p}{\bar{a}_{pk}}$$

We can just do a linear search over every i to calculate this

$$t_{max} = \min_{a_{ik} > 0} \left\{ \frac{\bar{b}_1}{\bar{a}_1}, \frac{\bar{b}_2}{\bar{a}_2}, \frac{\bar{b}_3}{\bar{a}_3}, \frac{\bar{b}_4}{\bar{a}_4} \right\} = \min_{a_{ik} > 0} \{1.9621, 3.0080, 4.7126\} = 1.9621$$

. So the index leaving is $\beta_p = \beta_1$.

Let $\nu_2 = 2$ and $\beta_1 = 3$ change place in **iteration 2**

Iteration 2

$$\begin{aligned}x^{(2)} &= \begin{pmatrix} 0 \\ 1.9621 \\ 0 \\ 1.0403 \\ 8.1771 \\ 8.0447 \end{pmatrix} \\r_{\nu}^{(2)} &= \begin{pmatrix} -3.9421 \\ 2.9778 \end{pmatrix} \\y^{(2)} &= \begin{pmatrix} -2.9778 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (2 \quad 4 \quad 5 \quad 6)\end{aligned}$$

Iteration 3

$$\begin{aligned}x^{(3)} &= \begin{pmatrix} 1.0745 \\ 3.0172 \\ 0 \\ 0 \\ 2.9652 \\ 7.0343 \end{pmatrix} \\r_{\nu}^{(3)} &= \begin{pmatrix} 4.0716 \\ -1.0919 \end{pmatrix} \\y^{(3)} &= \begin{pmatrix} 1.0919 \\ -4.0716 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (2 \quad 1 \quad 5 \quad 6)\end{aligned}$$

Iteration 4

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

$$r_\nu^* = \begin{pmatrix} 1.3634 \\ 0.5406 \end{pmatrix}$$

$$y^* = \begin{pmatrix} 0 \\ -1.3634 \\ -0.5406 \\ 0 \end{pmatrix}$$

$$\beta = (2 \quad 1 \quad 3 \quad 6)$$

All reduced costs are $r_\nu \geq 0$ so we found the optimal solution.

Optimal BFS

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

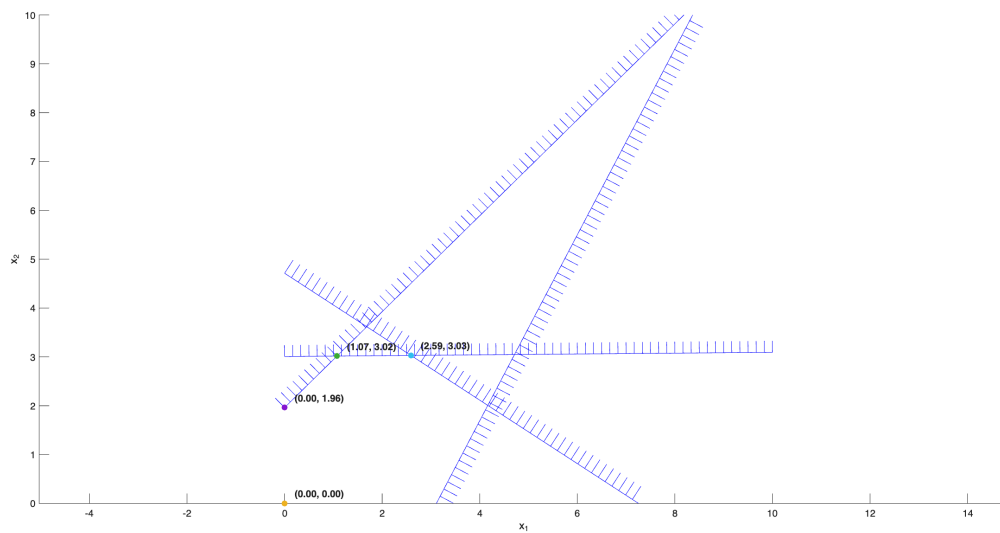


Figure 2: feasible region with iteration points included

Exercise 4

We'll repeat exercise 3 but with the choice of $q = 1$ in **iteration 1**. This is still valid since $r_{\nu_1} = -1.0324$ is also negative. $z = \bar{z} + r_{\nu}x_{\nu}$. If x_1 becomes a basic variable it still ensures the cost decreases in the first iteration (just less than if we chose x_2 first).

Iteration 1

$$\begin{aligned}x^{(1)} &= \begin{pmatrix} 0 \\ 0 \\ 1.9526 \\ 2.9920 \\ 14.0105 \\ 6.0423 \end{pmatrix} \\r_{\nu}^{(1)} &= \begin{pmatrix} -1.0324 \\ -2.9633 \end{pmatrix} \\y^{(1)} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 5 \quad 6)\end{aligned}$$

Iteration 2

$$\begin{aligned}x^{(2)} &= \begin{pmatrix} 3.1105 \\ 0 \\ 4.9920 \\ 3.0184 \\ 8.0035 \\ 0 \end{pmatrix} \\r_{\nu}^{(2)} &= \begin{pmatrix} 0.5315 \\ -3.5057 \end{pmatrix} \\y^{(2)} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ -0.5315 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 5 \quad 1)\end{aligned}$$

Iteration 3

$$\begin{aligned}x^{(3)} &= \begin{pmatrix} 4.1650 \\ 2.0071 \\ 4.0251 \\ 1.0310 \\ 0 \\ 0 \end{pmatrix} \\r_{\nu}^{(3)} &= \begin{pmatrix} -0.3425 \\ 0.8791 \end{pmatrix} \\y^{(3)} &= \begin{pmatrix} 0 \\ 0 \\ -0.8791 \\ 0.3425 \end{pmatrix} \\\beta &= (3 \quad 4 \quad 2 \quad 1)\end{aligned}$$

Iteration 4

$$\begin{aligned}x^* &= \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix} \\r_{\nu}^* &= \begin{pmatrix} 1.3634 \\ 0.5406 \end{pmatrix} \\y^* &= \begin{pmatrix} 0 \\ -1.3634 \\ -0.5406 \\ 0 \end{pmatrix} \\\beta &= (3 \quad 6 \quad 2 \quad 1)\end{aligned}$$

All reduced costs are $r_{\nu} \geq 0$, optimal solution found.

Optimal BFS

$$x^* = \begin{pmatrix} 2.5900 \\ 3.0302 \\ 1.4680 \\ 0 \\ 0 \\ 4.1036 \end{pmatrix}$$

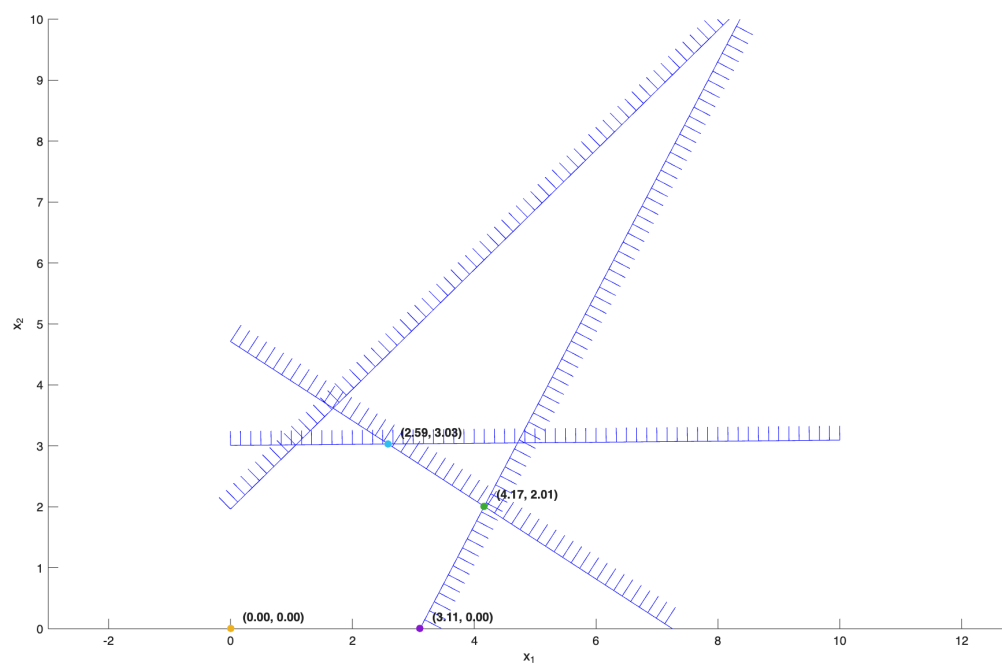


Figure 3: feasible region with iteration points included, but with choice of $q = 1$ in `it1`